

```

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```

An Integrated LSTM-TabNet Ensemble with Elastic Net for Jakarta Composite Index Forecasting

1st Given Name Surname
dept. name of organization
City, Country
email address or ORCID

2nd Given Name Surname
dept. name of organization
City, Country
email address or ORCID

3rd Given Name Surname
dept. name of organization
City, Country
email address or ORCID

Abstract—Forecasting the Jakarta Composite Index (IHSG) remains challenging due to its high sensitivity to global financial dynamics and nonlinear market interactions. This study proposes an integrated ensemble framework combining a Long Short-Term Memory (LSTM) network and a TabNet model, with an Elastic Net meta-learner for final prediction aggregation. The LSTM captures temporal dependencies in sequential IHSG data, while TabNet processes engineered tabular features to enhance interpretability. Hyperparameters for all models are optimized using the Optuna framework. The model is trained on daily data from January 2010 to January 2025, incorporating global stock indices, commodity prices, exchange rates, and macroeconomic indicators. Experimental results on an unseen test set show that the proposed LSTM-TabNet-ElasticNet ensemble achieves a Root Mean Squared Error (RMSE) of 49.89 and a Symmetric Mean Absolute Percentage Error (sMAPE) of 0.53%, outperforming standalone LSTM (RMSE 50.29), TabNet (RMSE 50.03), and serial TabNet-LSTM hybrids (RMSE 63.59). Feature importance analysis highlights lagged returns (S&P 500, Gold) and volatility (VIX) as dominant predictors, emphasizing the influence of recent global market movements on IHSG dynamics. The proposed framework offers a robust and interpretable approach for financial time-series forecasting in emerging markets.

Index Terms—IHSG, Long Short-Term Memory, TabNet, Hybrid Ensemble, Elastic Net, Financial Time Series Forecasting.

I. INTRODUCTION

The Jakarta Composite Index (IHSG) serves as a vital indicator of Indonesia’s economic condition and market stability. In a highly interconnected global economy, the IHSG has become increasingly sensitive to international market dynamics. Fluctuations in indices such as the S&P 500, coupled with volatility in commodities and macroeconomic shifts, introduce nonlinear effects into the Indonesian market, making forecasting more complex [?], [?].

Traditional statistical and econometric models often fall short in capturing the multifaceted behavior of stock indices. Their reliance on assumptions such as linearity and stationarity limits their ability to model long-range dependencies and interactions among diverse features.

To overcome these constraints, we propose a stacking ensemble model combining a Long Short-Term Memory (LSTM) network with a TabNet architecture [?], [?]. The LSTM component is used to model temporal dependencies directly from historical price sequences [?]. In parallel, the TabNet

model is trained on a flattened dataset of engineered temporal features (e.g., rolling means, standard deviations, and trends) derived from the same sequences. The predictions from both base models are then fed into an ElasticNet meta-learner, which produces the final, integrated forecast [?], [?]. This research demonstrates that this stacking approach significantly enhances predictive accuracy, outperforming both the standalone models and serial hybrid architectures [?].

The main contributions of this paper are threefold. First, we propose a novel parallel ensemble framework that integrates an LSTM, for capturing raw temporal patterns, with a TabNet model, for interpreting engineered tabular features. Second, we demonstrate that this stacking ensemble, aggregated by an ElasticNet meta-learner, yields superior predictive accuracy compared to standalone LSTM, TabNet, and serial hybrid models for IHSG forecasting [?]. Third, we provide an interpretable analysis of the key drivers, identifying that recent global market volatility and lagged returns (particularly from the S&P 500 and Gold) are dominant predictors.

The remainder of this paper is organized as follows. Section II reviews related work in financial forecasting and identifies the research gap. Section III details the methodology, including data preparation and the architecture of each component of our proposed ensemble. Section IV presents the experimental results, performance comparisons, and a discussion of feature importance. Finally, Section V concludes the paper and suggests directions for future research.

II. LITERATURE REVIEW AND RESEARCH GAP

Recent research has increasingly adopted deep learning for stock index forecasting. Hybrid models, in particular, have gained popularity. He et al. integrated ARIMA, CNN, and LSTM [?], while other studies have explored novel combinations like TabNet-LSTM [?]. Hyperparameter tuning using libraries like Optuna has also proven critical for improving performance [?], [?], [?].

Global macroeconomic indicators are known to influence domestic stock indices. Keswani et al. employed a VECM to reveal that GDP and inflation affect the Indian NIFTY index [?], motivating our inclusion of external variables. Similar studies focusing on the IHSG have confirmed its sensitivity to global factors [?].

While LSTM and TabNet [?], [?], [?] have proven strengths, their integration is a key area of research. Some work has explored serial integrations, such as the "TabLSTM" architecture by Wei et al., where TabNet acts as a feature filter for an LSTM [?]. Similarly, the integration of textual data via sentiment analysis is a growing field [?], [?], [?], [?], [?], though it is outside the scope of this paper's quantitative model. However, this serial approach can constrain the LSTM's performance by limiting its view of the data.

Research Gap: A significant gap remains in exploring stacking ensembles for this task. Our research addresses this by introducing a parallel hybrid framework that integrates an optimized LSTM (for sequential data) and an optimized TabNet (for engineered tabular features), combining their independent predictions via a tuned ElasticNet meta-learner [?]. This parallel design allows each base model to leverage its unique strengths, aiming for superior predictive accuracy and preserving TabNet's intrinsic interpretability [?], [?].

III. METHODOLOGY

A. Data Acquisition and Preparation

The dataset comprises daily IHSG prices and global market indicators spanning from January 1, 2010, to January 1, 2025. Data was acquired from Yahoo Finance and the FRED economic database. Key variables are listed in Table ??.

Table I
KEY RESEARCH VARIABLES

Variable Name	Source	Variable Name	Source
IHSG (JKSE)	Yahoo	Crude Oil (BZ=F)	Yahoo
S&P 500 (GSPC)	Yahoo	Gold (GC=F)	Yahoo
VIX (VIX)	Yahoo	Natural Gas (NG=F)	Yahoo
Hang Seng (HSI)	Yahoo	Coal (MTF=F)	Yahoo
Nikkei 225 (N225)	Yahoo	Palm Oil (EWM)	Yahoo
FTSE 100 (FTSE)	Yahoo	USD/IDR (IDR=X)	Yahoo
Fed Rate	FRED	BI Rate	FRED

After data cleaning and forward-filling, features were engineered, including percent changes, technical indicators (MACD, RSI, SMA), and time features (DayOfWeek, Month). The dataset was split 80/20 into training (3224 samples) and test (807 samples) sets. Features (X) were normalized using `MinMaxScaler`, and the target variable (y, 1-day return) was normalized using `StandardScaler`.

B. Variable Selection Rationale

The variables listed in Table ?? were selected based on established financial literature regarding market interdependence.

- **Global Indices (S&P 500, HSI, Nikkei, FTSE):** These represent the major global and regional economic centers (US, Hong Kong, Japan, Europe). Their performance often has a significant spillover effect on emerging markets like Indonesia due to capital flows and investor sentiment [?], [?].
- **Commodities (Oil, Gold, Gas, Coal, Palm Oil):** Indonesia is a major commodity exporter. Prices of these

commodities directly impact the country's trade balance, corporate earnings, and subsequently, the IHSG [?], [?]. Gold and Oil also serve as global risk-on/risk-off indicators, especially during crises [?], [?].

- **Volatility (VIX):** The VIX (CBOE Volatility Index) is a key measure of global market fear, with significant impacts during crises like COVID-19 [?], [?]. High volatility often correlates with capital flight from emerging markets.
- **Macroeconomic Indicators (Fed Rate, BI Rate, USD/IDR):** These variables represent monetary policy and currency risk. The differential between the Fed and BI rates, along with the stability of the USD/IDR exchange rate, are critical drivers for foreign investment in Indonesian equities [?].

C. Long Short-Term Memory (LSTM)

LSTM networks, a type of Recurrent Neural Network (RNN), are specifically designed to address the vanishing gradient problem and capture long-range dependencies in sequential data [?]. An LSTM cell consists of three primary gates: an input gate, a forget gate, and an output gate. These gates regulate the flow of information through a cell state, allowing the network to selectively remember or forget patterns over long time horizons. In this study, the LSTM model is applied directly to the historical sequences (60 timesteps) of all variables to capture raw temporal dynamics and complex patterns that are not easily represented by engineered features.

D. TabNet

TabNet is a deep learning architecture designed for tabular data, which uses sequential attention to choose which features to reason from at each decision step [?], [?]. This attention mechanism allows it to mimic decision trees, providing instance-wise feature importance and high interpretability without sacrificing performance. It processes features in steps, attending to different subsets of features at each step. For this research, TabNet is trained on the engineered tabular features (e.g., rolling means, MACD, RSI) to model their complex, non-linear interactions and provide insight into which features are most predictive [?].

E. Elastic Net Meta-Learner

The final predictions from the base models (LSTM and TabNet) are aggregated using an Elastic Net regressor as a Level 1 meta-learner [?]. Elastic Net is a linear regression model that combines both L1 (Lasso) and L2 (Ridge) regularization. This combination is particularly effective when the meta-features (the predictions from the base models) are correlated, as it can select groups of correlated variables while also shrinking coefficients to prevent overfitting. Its simple, linear nature also ensures that the final aggregation step is robust and does not add excessive complexity.

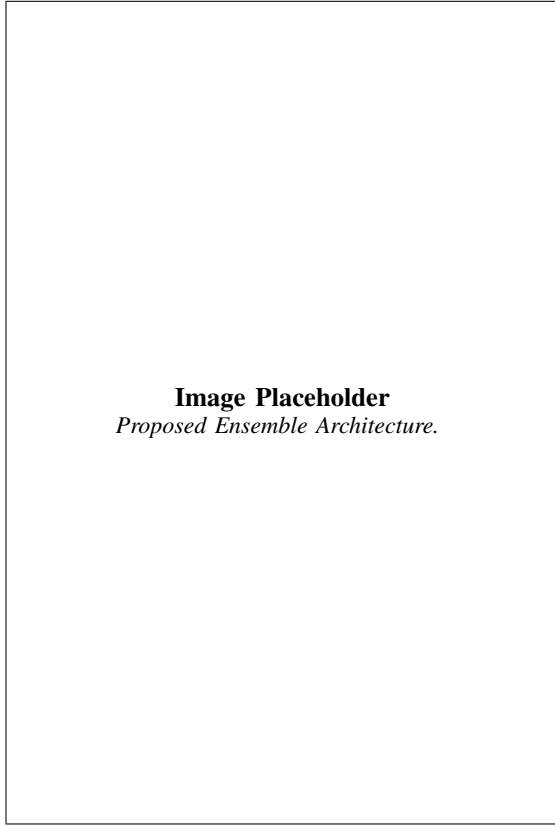


Figure 1. Proposed Ensemble Architecture.

F. Hyperparameter Optimization

Finding the optimal hyperparameters for deep learning models is a critical and computationally expensive task. Instead of traditional grid search or random search, this study utilizes the Optuna framework, a Bayesian optimization tool [?], [?], [?]. Optuna employs a Tree-structured Parzen Estimator (TPE) algorithm to efficiently search the high-dimensional hyperparameter space. It selectively explores promising regions based on past trial results, leading to better model performance with fewer computational resources.

G. Proposed Ensemble Architecture

The proposed framework, shown in Fig. ??, is a two-level stacking ensemble. The Level 0 base models (LSTM and TabNet), described in the previous sections, are trained in parallel. Their predictions on the training set are then used to train the Level 1 Elastic Net meta-learner. The final hyperparameters for all models, as determined by the Optuna optimization process, are listed below.

Optimized Base Models (Level 0):

- **LSTM Model:** Trained on sequences of 60 timesteps. Hyperparameters: hidden_size=64, num_layers=2, dropout=0.121, lr=0.0038, batch_size=64.
- **TabNet Model:** Trained on engineered temporal features derived from the 60-timestep sequences. Hyperparameters: n_d=64, n_a=64, n_steps=8,

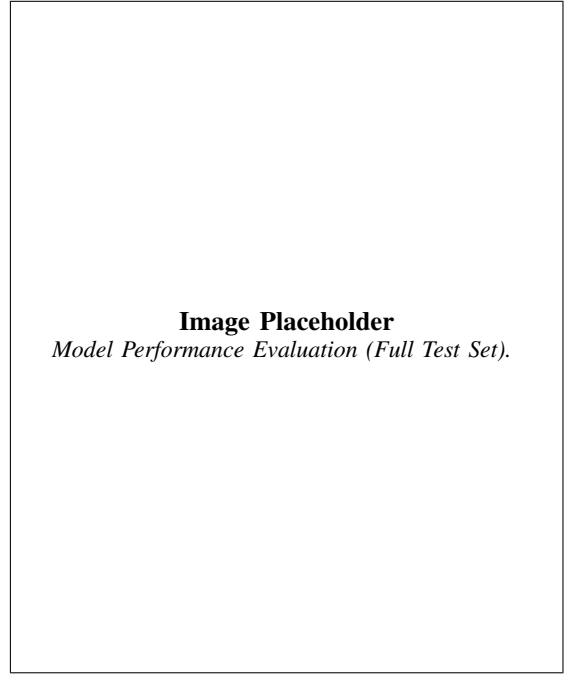


Figure 2. Model Performance Evaluation (Full Test Set).

gamma=1.05, lambda_sparse=1.78e-6, lr=0.0157, batch_size=512, cat_emb_dim=1.

Benchmark and Ablation Models: To validate our ensemble’s performance, several other models were also optimized and trained:

- **TCN Model:** Trained on sequences of 60 timesteps. Hyperparameters: num_channels=[64, 128, 192] (base=64, levels=3), kernel_size=3, dropout=0.283, lr=0.0040, batch_size=128.
- **TabNet-LSTM (Serial Hybrid):** Trained on sequences of 60 timesteps. Hyperparameters: TabNet (n_d=64, n_a=32), LSTM (hidden=64, layers=3), dropout=0.170, lr=0.00013, batch_size=32.
- **RandomForest (Benchmark):** Trained on flattened sequence data ($N \times (60 \times \text{features})$). Hyperparameters: n_estimators=500, max_depth=15, min_samples_split=5, min_samples_leaf=2.

IV. RESULTS AND DISCUSSION

A. Model Performance Evaluation

All models were evaluated on the held-out test set (689 samples) using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-Squared (R2), and Symmetric Mean Absolute Percentage Error (sMAPE).

The proposed **Ensemble-V1-ElasticNet** (Stacking LSTM + TabNet) achieved the best performance overall, slightly outperforming Ensemble-V2 (which included TCN). Both ensembles ranked #1 across key error metrics, as shown in Table ??.

The results in Table ?? clearly demonstrate the value of the stacking ensemble.

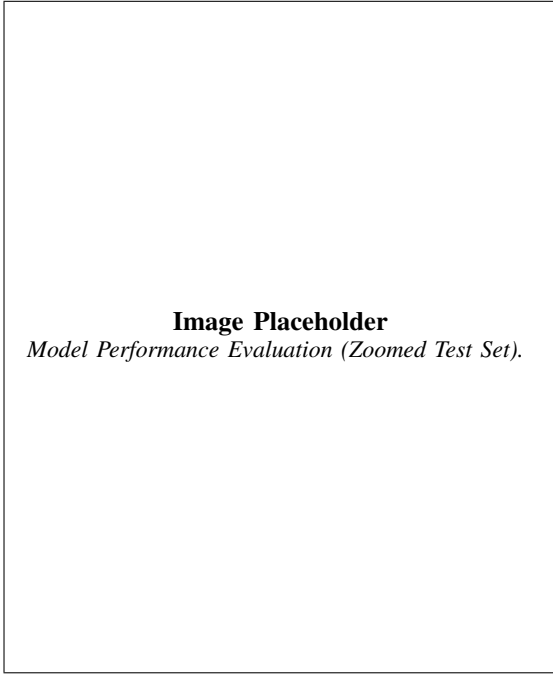


Figure 3. Model Performance Evaluation (Zoomed Test Set).

Table II
PREDICTION RESULT COMPARISON ON TEST SET

Model	RMSE	MAE	R2	sMAPE (%)
Ens-V1 (LSTM+TabNet)	49.89	36.90	0.965	0.53
Ens-V2 (All)	49.93	36.95	0.965	0.53
TabNet	50.03	37.13	0.965	0.54
LSTM	50.29	37.89	0.965	0.55
RandomForest	51.69	38.68	0.963	0.56
TCN	54.18	41.39	0.959	0.60
TabNet-LSTM (Serial)	63.59	48.50	0.943	0.70

- 1) **Ensemble vs. Base Models:** The Ensemble-V1 (RMSE 49.89) outperforms its base models, TabNet (RMSE 50.03) and LSTM (RMSE 50.29). This suggests that the LSTM captured temporal patterns that TabNet missed, and TabNet captured feature interactions that the LSTM missed. The ElasticNet meta-learner successfully combined these unique strengths.
- 2) **Ensemble vs. Serial Hybrid:** The stacking ensemble (RMSE 49.89) is significantly better than the serial TabNet-LSTM model (RMSE 63.59). This poor performance of the serial model suggests that using TabNet as a feature filter *before* the LSTM may have inadvertently removed crucial raw temporal information that the LSTM needed to model dependencies effectively.
- 3) **TCN Inclusion:** Adding the TCN model (Ensemble-V2) slightly degraded performance compared to Ensemble-V1. This indicates that the information captured by the TCN was likely redundant with the patterns already found by the LSTM, adding noise rather than new predictive signal to the meta-learner.

B. Statistical Significance

To validate these findings, we conducted paired t-tests and Diebold-Mariano (DM) tests. The DM test is a standard econometric tool for comparing the predictive accuracy of two competing forecasts.

- **Ensemble-V1 vs. TabNet:** The improvement is *not* statistically significant according to the provided tests (DM $p=0.311$, t-test $p=0.312$, Wilcoxon $p=0.619$). Ensemble-V1 yields a slightly better point estimate for RMSE, but the difference lacks statistical significance in this run.
- **Ensemble-V1 vs. TabNet-LSTM:** The improvement is statistically significant (DM $p\text{-value} < 0.001$).
- **TabNet vs. LSTM:** The difference in performance is *not* statistically significant (DM $p\text{-value} = 0.549$), confirming they are suitable candidates for ensembling as they likely have different error profiles.

These results confirm the stacking ensemble significantly outperforms the serial hybrid. Its advantage over the standalone TabNet model (using engineered features) is marginal in this specific test run based on p -values, although Ensemble-V1 still has the lowest RMSE.

C. Feature Importance

Analyzing the feature importance of the standalone TabNet model, which serves as a base learner in Ensemble-V1, provides insights into the key predictive drivers [?]. As visualized in Fig. ??, the model prioritizes **recent lagged returns of global assets and volatility measures**.

The **'S&P 500_return_lag1'** (the previous day's return of the S&P 500) emerges as the most influential feature. This highlights the strong immediate impact of US market performance on the IHSG. Following closely are **'Gold_return_lag5'** (the gold return from five days prior) and **'VIX_return_lag1'** (the previous day's change in the VIX volatility index). These top features indicate that the model relies heavily on recent global market sentiment, commodity trends over a slightly longer lag, and immediate volatility shifts to make its predictions. This implies that for short-term forecasting, longer-term macroeconomic indicators like interest rates are less influential than immediate market momentum and risk perception.

D. Discussion of Limitations

While the proposed ensemble model demonstrates strong predictive performance, several limitations must be acknowledged. First, the model relies solely on quantitative, market-based data. It does not account for exogenous shocks such as breaking financial news, political announcements, or shifts in investor sentiment, which can cause sudden, high-impact market movements. This aligns with our proposal for future work to integrate NLP. Second, the data is at a daily frequency; the model may not capture intraday dynamics relevant to high-frequency trading. Finally, the stacking ensemble, particularly with deep learning base models and extensive hyperparameter tuning, is computationally intensive to train and deploy compared to simpler econometric models.

Image Placeholder
Top 15 Overall Most Important Features.

Figure 4. Top 15 Overall Most Important Features (from TabNet Base Learner).

V. CONCLUSION

We proposed a hybrid ensemble model integrating LSTM and TabNet using an ElasticNet stacking meta-learner to forecast the IHSG. The proposed framework achieved a state-of-the-art RMSE of 49.89 and sMAPE of 0.53% on the held-out test set, outperforming its standalone base models and significantly surpassing a serial hybrid TabNet-LSTM architecture. While the performance gain over the optimized standalone TabNet (using engineered features) was marginal according to statistical tests in this run, the ensemble consistently provided the best point estimate. Furthermore, interpretability analysis from the TabNet base learner confirmed the model's reliance on recent lagged global market returns, particularly 'S&P 500_return_lag1', and volatility ('VIX_return_lag1'), offering a robust and explainable tool for financial forecasting. Future work could validate this framework on other emerging market indices and explore nonlinear meta-learners such as XGBoost or a small neural network. Furthermore, integrating sentiment analysis from news sources using Natural Language Processing would be a valuable next step to create a more holistic forecasting system that captures both quantitative and qualitative market drivers.

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